

Late Breaking Results: Extracting QNNs from NISQ Computers via Ensemble Learning

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ABSTRACT

The recent success of Quantum Neural Networks (QNNs) prompts model extraction attacks on cloud platforms, even under black-box constraints. These attacks repeatedly query the victim QNN with malicious inputs. However, existing extraction attacks tailored for classical models yield local substitute QNNs with limited performance due to NISQ computer noise. Drawing from bagging-based ensemble learning, which uses independent weak learners to learn from noisy data, we introduce a novel QNN extraction approach. Our experimental results show this quantum ensemble learning approach improves local QNN accuracy by up to 15.09% compared to previous techniques.

KEYWORDS

Model Extraction Attack, Quantum Neural Networks, Variational Quantum Circuits, NISQ

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1 INTRODUCTION

Motivation. Quantum Neural Networks (QNNs) utilize Variational Quantum Circuits (VQCs)[7], iteratively trained on datasets, for diverse tasks[1, 2, 4]. Developing accurate VQCs demands expertise and resources, commonly deployed on cloud-based Noisy Intermediate Scale Quantum (NISQ) machines for accessibility. Due to the significant investment, VQCs represent valuable intellectual property (IP) vulnerable to model extraction attacks[3, 8, 11, 12], where adversaries aim to replicate QNN behavior by querying it multiple times to create a new dataset for training local substitute QNNs.

Existing model extraction [3, 8, 11, 12] are designed for classical systems and overlook quantum noise. In our preliminary study, we utilized a state-of-the-art (SOTA) QNN [10] to train a 2-layer victim QNN on the cloud. Employing the SOTA extraction method, *CloudLeak*[12], we constructed a substitute QNN. As shown in Figure 1, *CloudLeak* replicates a substitute QNN with a 5% to 10% reduction in accuracy compared to the victim QNN under ideal conditions. However, evaluating the learned QNN on the NISQ device IBM_Auck1and revealed a significant accuracy decrease. This

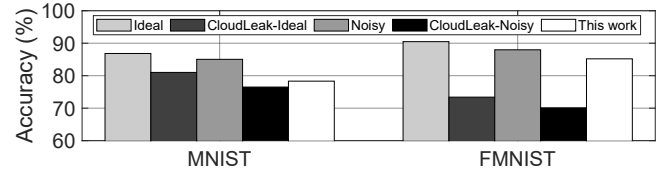


Figure 1: The accuracy of victim QNN and CloudLeak attacks [12].

decline primarily stems from inherent quantum noise in NISQ [9], introducing noisy labels during queries to the cloud model.

Contributions. This paper explores QNN model extraction from NISQ computers through an ensemble of local QNN learners, drawing inspiration from classical ensemble learning methodologies [6]. We specifically employ a bagging-based ensemble construction [5], which assembles a committee of independent weak learners and employs decision fusion for prediction, known for its efficacy in handling noisy data compared to alternatives like boosting and randomization. Experimental results on MNIST and Fashion-MNIST datasets illustrate that our method enhances local QNN accuracy by up to 15.09% compared to existing classical model stealing techniques, as shown in the rightmost bar of Figure 1.

2 EXTRACTING QNN VIA ENSEMBLE LEARNING

Threat Model & Problem Formulation. This paper operates under a practical threat model where the adversary interacts with the victim QNN as a black box, lacking access to internal details such as circuit architecture, training data, and noise levels on the NISQ machine. Through the collection of input-output pairs, the adversary trains a local substitute QNN to emulate the performance of the cloud-based QNN, thus enabling unlimited query tasks without cost. This process generally involves three key steps: (1) Initiation of Substitute QNN: The adversary initializes a basic substitute local QNN model, denoted as Q_A , chosen from a function family $\mathcal{Q}$, and sets its hyperparameters. (2) Victim QNN Query: The adversary gathers an initial set of unlabeled samples, referred to as S , by querying the victim model Q_V and records the responses in a dataset $D = S, Q_V(S)$. (3) Substitute Model Training: The adversary proceeds to train the substitute QNN Q_A with the dataset D , employing specific training strategies, optimizers, and decision-fusion schemes. In the following, we introduce our methodology for QNN model extraction based on ensemble learning, providing comprehensive explanations for the configuration of each step.

Step1: Initiating Substitute Model. Unlike classical machine learning, contemporary QNNs follow a standardized structure, typically relying on a VQC framework. This framework commonly integrates a rotation layer with 1-Qubit gates and entanglement layers utilizing 2-Qubit gates arranged in a ring structure. In our investigation, we assess three prominent QNN models, denoted as A1 [10], A2 [1], and A3 [2], which serve as the pool of candidate QNN models. Both the victim QNN deployed on the cloud and the local substitution

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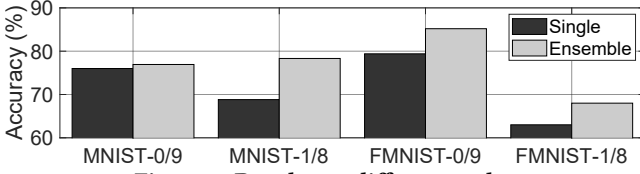


Figure 2: Results on different tasks.

model are chosen from this model pool. Additionally, we aim to illustrate the influence of the substitute QNN's structure on the final accuracy of the learned model. Once a local QNN substitute model is selected, we establish an ensemble of instance QNNs as committed classifiers. In this initial exploration, we maintain the committee number at 5. Each of these QNN instances undergoes individual training using different bootstrap sub-datasets through bagging.

Step2: Querying the Victim QNN. In our attack methodology, akin to classical model stealing techniques such as Cloudleak, a pivotal phase involves querying the victim model. Initially, we submit unlabeled data samples to the victim hosted in the cloud to gather predictions. These predictions, along with the original unlabeled samples, form an integrated training dataset. It's essential to note that predictions from the VQC generally emerge as probability vectors, with each element representing the likelihood of a specific category. Subsequently, we assign single-category labels based on these probability vectors, thereby creating the final training datasets by combining the samples and obtained labels. To address the variability of quantum noise in cloud-based victim models over time, we strategically query the victim at different intervals throughout the day, specifically every 24/ m hours, to better understand the behavior of the victim model. In our preliminary experiments, we set m to 4.

Step3: Training the Local Substitute Model. Upon obtaining the comprehensive training dataset from the victim, we utilize the Bagging method to generate distinct sub-training sets, which are then employed to train various sub-VQCs within the ensemble committee. Specifically, the bagging technique entails repeatedly sampling from the comprehensive training dataset with replacement to create new sub-training sets. This iterative process generates diverse sub-training sets, effectively introducing variability among the sub-VQCs and mitigating overfitting to noisy labels [5]. We employ the Negative Log-Likelihood (NLL) loss function, commonly utilized in classification tasks, during the training process, utilizing the sample-single label dataset as mentioned earlier.

In exploring decision-making strategies within the ensemble, we evaluate various fusion schemes including unweighted averaging, majority vote, and weighted methods based on individual influence. Through careful examination, we observe no discernible differences among the candidate methods. Consequently, we opt for the majority vote as the default fusion strategy due to its simplicity of implementation.

3 EVALUATION

Experimental Setup. We conduct experiments utilizing the MNIST and Fashion-MNIST datasets. All images undergo downsampling from the size 28×28 to 1×8 via principle component analysis. For MNIST, our focus lies on classifying between (i) 0&9 and (ii) 1&8; whereas for Fashion-MNIST, we concentrate on 2-category classification: (i) T-shirts&boots and (ii) Trousers&bags. The experimental setups are executed using Qiskit and the IBM Quantum System backend, specifically configured with parameters for the 27-qubit

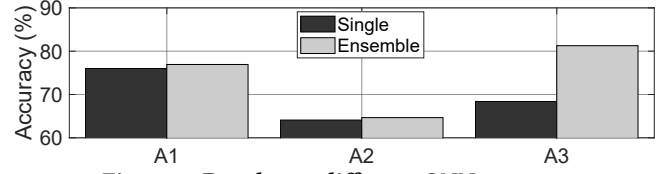


Figure 3: Results on different QNN ansatze.

IBM_Auckland. We query 3000 images for each task and reserve another 1000 images for the test dataset.

Overall Performance. Our approach improves the substitute QNN accuracy on NISQ computers by up to 15.09% as shown in Figure 1. We further compare ensemble learning-based methods to a baseline where only a single local QNN substitute model is trained. Figure 2 illustrates a significant accuracy improvement with ensemble learning-based model extraction. Across various task scenarios, the substitute networks trained in this study achieve performance comparable to the victim VQC, highlighting the effectiveness of ensemble learning in QNN training, particularly in scenarios with noisy data.

Impact of VQC Ansatz. As depicted in Figure 3, while the ensemble learning-based approach enhances performance, variations are evident across different VQC ansatze. Notably, A3 consistently outperforms others, attributed to its ability to dynamically adapt and learn task-specific entanglement through parameterized two-qubit gates. These findings underscore the significant impact of model structure choice on the efficacy of the QNN model extraction attack.

4 CONCLUSION

In this paper, we introduce a novel model extraction attack for Quantum Neural Networks on contemporary noisy cloud computers. Our experimental results demonstrate that this quantum ensemble learning approach enhances local QNN accuracy by up to 15.09% compared to previous techniques.

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